```
# Install required libraries (RUN THIS FIRST)
!pip install transformers sentencepiece torch --quiet

# Import libraries
from transformers import pipeline
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

class Translator:
    def __init__(self, model_name):
        self.tokenizer = AutoTokenizer.from_pretrained(model_name)
        self.model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

    def __call__(self, text):
        inputs = self.tokenizer(text, return_tensors="pt", padding=True, truncation=True)
        translated_tokens = self.model.generate(**inputs, max_length=512)
        return self.tokenizer.decode(translated_tokens[0], skip_special_tokens=True)

# English → French
translator_en_fr = Translator("Helsinki-NLP/opus-mt-en-fr")

# English → German
translator_en_de = Translator("Helsinki-NLP/opus-mt-en-de")

# English → Hindi
translator_en_hi = Translator("Helsinki-NLP/opus-mt-en-hi")

print("Models loaded successfully and manual translation wrappers created!")
```

```
/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:176: UserWarning: Recommended: pip
  warnings.warn("Recommended: pip install sacremoses.")
Loading weights: 100%  258/258 [00:00<00:00, 495.54it/s, Materializing param=model.shared.weight]
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight
config.json: 1.33k/? [00:00<00:00, 73.6kB/s]
tokenizer_config.json: 100%  42.0/42.0 [00:00<00:00, 4.17kB/s]
source.spm: 100%  768k/768k [00:00<00:00, 45.2MB/s]
target.spm: 100%  797k/797k [00:00<00:00, 41.4MB/s]
vocab.json: 1.27M/? [00:00<00:00, 26.7MB/s]
pytorch_model.bin: 100%  298M/298M [00:06<00:00, 47.3MB/s]
Loading weights: 100%  258/258 [00:00<00:00, 541.54it/s, Materializing param=model.shared.weight]
model.safetensors: 100%  298M/298M [00:04<00:00, 149MB/s]
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight
generation_config.json: 100%  293/293 [00:00<00:00, 14.5kB/s]
config.json: 1.39k/? [00:00<00:00, 26.9kB/s]
tokenizer_config.json: 100%  44.0/44.0 [00:00<00:00, 963B/s]
source.spm: 100%  812k/812k [00:00<00:00, 6.50MB/s]
target.spm: 100%  1.07M/1.07M [00:00<00:00, 9.94MB/s]
vocab.json: 2.10M/? [00:00<00:00, 19.0MB/s]
pytorch_model.bin: 100%  306M/306M [00:02<00:00, 226MB/s]
Loading weights: 100%  258/258 [00:00<00:00, 669.62it/s, Materializing param=model.shared.weight]
model.safetensors: 0%  0.00/306M [00:00<?, ?B/s]
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight
generation_config.json: 100%  293/293 [00:00<00:00, 21.5kB/s]
Models loaded successfully and manual translation wrappers created!
```

```
sentences = [
    # Simple
    "I am happy.",
    "She is reading a book.",

    # Compound
    "I wanted to go outside, but it was raining.",
    "He studied hard, so he passed the exam.",
```

```
    # Complex
    "Although it was late, she continued working.",
    "If you practice daily, you will improve.",

    # Interrogative
    "What is your name?",
    "Can you solve this problem?",

    # Technical
    "Machine learning is a subset of artificial intelligence.",
    "Cybersecurity protects systems from digital attacks.",

    # Conversational
    "Hey, what's going on?",
    "Let's meet tomorrow at the cafe.",

    # News/Academic
    "The government announced new policies today.",
    "Climate change is affecting global temperatures.",

    # Long sentence
    "Even though the project was difficult, the team managed to complete it successfully on time."
]
```

```
print("----- TRANSLATION OUTPUT -----\n")

for sentence in sentences:
    fr = translator_en_fr(sentence)
    de = translator_en_de(sentence)
    hi = translator_en_hi(sentence)

    print("English :", sentence)
    print("French  :", fr)
    print("German  :", de)
    print("Hindi   :", hi)
    print("-" * 60)
```

```
----- TRANSLATION OUTPUT -----

English : I am happy.
French  : Je suis heureuse.
German  : Ich bin glücklich.
Hindi   : मैं खुश हूँ.
------------------------------------------------------------
English : She is reading a book.
French  : Elle lit un livre.
German  : Sie liest ein Buch.
Hindi   : वह एक किताब पढ़ रहा है.
------------------------------------------------------------
English : I wanted to go outside, but it was raining.
French  : Je voulais sortir, mais il pleuvait.
German  : Ich wollte rausgehen, aber es regnete.
Hindi   : मैं बाहर जाना चाहता था, लेकिन यह बारिश थी.
------------------------------------------------------------
English : He studied hard, so he passed the exam.
French  : Il a étudié dur, donc il a réussi l'examen.
German  : Er hat hart studiert, also hat er die Prüfung bestanden.
Hindi   : उसने कठिन अध्ययन किया, सो वह परीक्षा से गुज़र गया ।
------------------------------------------------------------
English : Although it was late, she continued working.
French  : Bien qu'il soit tard, elle a continué à travailler.
German  : Obwohl es spät war, arbeitete sie weiter.
Hindi   : हालाँकि देर हो चुकी थी, फिर भी वह काम करती रही ।
------------------------------------------------------------
English : If you practice daily, you will improve.
French  : Si vous pratiquez quotidiennement, vous vous améliorerez.
German  : Wenn Sie täglich üben, werden Sie sich verbessern.
Hindi   : अगर आप रोज़ कसरत करते हैं, तो आप सुधार पाएँगे ।
------------------------------------------------------------
English : What is your name?
French  : Quel est votre nom ?
German  : Wie heißt du?
Hindi   : आपका Windows Live कूटशब्द क्या है?
------------------------------------------------------------
English : Can you solve this problem?
French  : Pouvez-vous résoudre ce problème ?
German  : Können Sie dieses Problem lösen?
Hindi   : क्या आप इस समस्या को हल कर सकते हैं?
------------------------------------------------------------
English : Machine learning is a subset of artificial intelligence.
French  : L'apprentissage automatique est un sous-ensemble de l'intelligence artificielle.
German  : Maschinelles Lernen ist eine Teilmenge künstlicher Intelligenz.
Hindi   : मशीन सीखने के लिए कृत्रिम बुद्धि की एक नक़ल है।
------------------------------------------------------------
English : Cybersecurity protects systems from digital attacks.
```

```
French  : La cybersécurité protège les systèmes contre les attaques numériques.
German  : Cybersecurity schützt Systeme vor digitalen Angriffen.
Hindi   : सी. बी.)
------------------------------------------------------------
English : Hey, what's going on?
French  : Qu'est-ce qui se passe ?
German  : Hey, was ist los?
Hindi   : अरे, क्या हो रहा है?
------------------------------------------------------------
English : Let's meet tomorrow at the cafe.
```

```
complex_sentences = [
    "Although the weather was bad, we decided to continue our journey.",
    "Artificial intelligence, which is rapidly evolving, is transforming industries.",
    "If you do not follow the rules, you may face consequences.",
    "What are the major challenges in natural language processing?",
    "Data science plays a crucial role in decision-making processes."
]

print("----- COMPLEX SENTENCES -----\n")

for sentence in complex_sentences:
    fr = translator_en_fr(sentence)
    de = translator_en_de(sentence)
    hi = translator_en_hi(sentence)

    print("English :", sentence)
    print("French  :", fr)
    print("German  :", de)
    print("Hindi   :", hi)
    print("-" * 60)
```

```
----- COMPLEX SENTENCES -----

English : Although the weather was bad, we decided to continue our journey.
French  : Bien que le temps ait été mauvais, nous avons décidé de poursuivre notre voyage.
German  : Obwohl das Wetter schlecht war, beschlossen wir, unsere Reise fortzusetzen.
Hindi   : हालाँकि मौसम खराब था, फिर भी हमने अपना सफर जारी रखने का फैसला किया ।
------------------------------------------------------------
English : Artificial intelligence, which is rapidly evolving, is transforming industries.
French  : L'intelligence artificielle, qui évolue rapidement, transforme les industries.
German  : Künstliche Intelligenz, die sich rasch entwickelt, verändert die Industrie.
Hindi   : कलात्मक बुद्धि, जो तेज़ी से बढ़ रही है, उद्योगों को परिवर्तित कर रही है ।
------------------------------------------------------------
English : If you do not follow the rules, you may face consequences.
French  : Si vous ne suivez pas les règles, vous pouvez faire face à des conséquences.
German  : Wenn Sie sich nicht an die Regeln halten, können Sie mit Konsequenzen rechnen.
Hindi   : अगर आप नियम नहीं मानते, तो आप शायद परिणाम भुगतें ।
------------------------------------------------------------
English : What are the major challenges in natural language processing?
French  : Quels sont les principaux défis dans le traitement du langage naturel?
German  : Was sind die großen Herausforderungen in der natürlichen Sprachverarbeitung?
Hindi   : प्राकृतिक भाषा की रचना में कौन - सी सबसे बड़ी चुनौतियाँ हैं?
------------------------------------------------------------
English : Data science plays a crucial role in decision-making processes.
French  : La science des données joue un rôle crucial dans les processus décisionnels.
German  : Die Datenwissenschaft spielt bei Entscheidungsprozessen eine entscheidende Rolle.
Hindi   : डेटा विज्ञान निर्णय करने की प्रक्रियाओं में एक अहम भूमिका अदा करता है।
------------------------------------------------------------
```